

Adaptive Hybrid Continuous Behavioral Authentication System Based on Keyboard and Mouse Dynamics for Real-Time User Verification

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Abstract—The rapid proliferation of cloud-based services and digitally interconnected systems has rendered traditional, static authentication paradigms fundamentally inadequate in the face of increasingly sophisticated cyber threats. Conventional usernamepassword mechanisms and even Multi-Factor Authentication (MFA) operate on a point-of-entry verification model, leaving active sessions vulnerable to session hijacking, credential theft, and automated bot attacks. This paper presents the design, implementation, and empirical evaluation of an Adaptive Hybrid Continuous Behavioral Authentication System that operationalizes the Zero-Trust security paradigm through persistent, real-time biometric analysis. The system continuously monitors keystroke dynamics and mouse movement patterns, extracting high-dimensional behavioral feature vectors for anomaly detection via the Isolation Forest algorithm. Empirical evaluation demonstrates a keystroke bot detection rate of 96.4% with a false positive rate of 3.2%, and a mouse behavior detection rate of 94.2% with a false positive rate of 4.1%, with inference latencies below 100 milliseconds.

Index Terms—Behavioral Biometrics, Continuous Authentication, Zero-Trust Architecture, Keystroke Dynamics, Mouse Dynamics, Isolation Forest, Anomaly Detection, Cybersecurity, Machine Learning, Real-Time Verification.

I. INTRODUCTION

IN an era characterized by unprecedented digital transformation and the exponential growth of cloud-based services, the imperative to secure sensitive data and user identities has never been more critical. The traditional paradigm of cybersecurity, which heavily relied on perimeter defense, has been rendered largely obsolete by sophisticated cyber threats, distributed workforces, and the proliferation of automated attack vectors. At the core of digital security lies authentication — the mechanism by which a system verifies the identity of a user.

Conventional authentication modalities are fundamentally static. They operate on a one-time verification principle, validating the user's identity only at the initial point of entry. Once access is granted, the system inherently trusts the active session until explicitly terminated. This “trust but verify once” approach creates a massive vulnerability window. If an adversary successfully compromises a session through session hijacking, cookie theft, or stolen credentials obtained via phishing, the system remains entirely oblivious to the intrusion.

Furthermore, the rise of sophisticated botnets and automated credential-stuffing tools has exponentially increased the ease with which traditional authentication barriers can be breached. According to the OWASP Top 10 (2021), broken access control and identification failures remain among the most critical web application security risks globally, underscoring the urgency for more robust, adaptive authentication mechanisms [1].

To mitigate these vulnerabilities, the cybersecurity industry has increasingly pivoted toward the Zero-Trust architecture, which operates on the foundational principle of “never trust, always verify.” The National Institute of Standards and Technology (NIST) Special Publication 800-207 formally defines Zero-Trust as a set of evolving cybersecurity paradigms that move defenses from static, network-based perimeters to focus on users, assets, and resources [2].

Continuous authentication leverages Behavioral Biometrics — the analysis of a user's unique physiological and cognitive patterns during their interaction with a device. By continuously monitoring metrics such as keystroke dynamics and mouse movement patterns, a system can persistently validate the

user's identity throughout the entirety of their session. This project proposes a robust, machine-learning-driven behavioral authentication system that operates imperceptibly in the background, ensuring high security without degrading the user experience.

The remainder of this paper is organized as follows: Section II reviews related literature. Section III details the system architecture and methodology. Section IV presents implementation and feature engineering. Section V provides testing, evaluation, and results. Section VI concludes the paper and outlines future work.

II. RELATED WORK

A. Evolution of Authentication Systems

The history of access control and user authentication is a chronicle of the perpetual arms race between security architectures and malicious actors. Knowledge-based authentication — the username and password combination — was first implemented on the Compatible Time-Sharing System (CTSS) at MIT in 1961 [3]. While passwords provided a scalable solution for decades, their efficacy has been systematically undermined by human cognitive limitations and the exponential increase in computational power available to attackers.

Data breach statistics underscore this vulnerability. The IBM Cost of a Data Breach Report consistently identifies stolen or compromised credentials as the leading initial attack vector globally [4]. To fortify single-factor authentication, the industry widely adopted Multi-Factor Authentication (MFA). Despite significantly elevating security posture, traditional MFA remains a static, point-of-entry defense mechanism, catalyzing the exploration of continuous and adaptive authentication mechanisms.

B. The Paradigm of Continuous Authentication and Zero-Trust

The limitations of static authentication catalyzed the transition toward the Zero-Trust security model. Coined by Forrester Research analyst John Kindervag in 2010, Zero-Trust operates on the foundational mandate: “Never trust, always verify” [5]. NIST formally codified this in Special Publication 800-207 (2020), defining Zero-Trust as an evolving set of cybersecurity paradigms that move defenses from static, network-based perimeters to focus on users, assets, and resources [2].

Continuous Authentication is the operationalization of the Zero-Trust philosophy at the user identity level. Unlike point-of-entry verification, continuous authentication systems monitor the user implicitly and persistently throughout the entire session, dynamically triggering graduated security responses upon detecting deviation from established behavioral patterns [6], [7].

C. Behavioral Biometrics

Behavioral Biometrics analyze the unique cognitive and motor skills exhibited by an individual while interacting with a computational system — encompassing keystroke dynamics, mouse dynamics, touchscreen gesture patterns, and gait analysis. The scientific foundation lies in the observation that these behaviors are shaped by individualized neuromuscular pathways and cognitive habits developed over years of device interaction, making them uniquely personal, highly stable, and extremely difficult to consciously replicate [8].

D. Anomaly Detection in Cybersecurity

The Isolation Forest algorithm proposed by Liu, Ting, and Zhou (2008) has demonstrated superior performance characteristics for high-dimensional behavioral biometric data [9]. Unlike distance-based methods, Isolation Forest directly targets anomalies by exploiting their defining characteristics — anomalies are few in number and different in value from normal instances — achieving near-linear time complexity during training and constant time during inference.

E. Gaps in Current Commercial Solutions

Commercial behavioral authentication solutions such as BioCatch, BehavioSec, and NeuroID exhibit critical limitations: prohibitive costs, proprietary SaaS architectures transmitting sensitive biometric data to third-party servers, batchprocessing approaches incompatible with real-time enforcement, and unaddressed cold-start vulnerabilities. This project addresses all identified gaps by architecting an open, selfhosted behavioral authentication system with real-time sub100ms processing and synthetic data generation for immediate cold-start resilience [10].

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Architectural Overview

The proposed system is engineered utilizing a modern client-server architecture, cleanly separating the user interface and biometric data collection layer (Frontend) from the core business logic, machine learning inference, and data persistence layer (Backend). The overall architecture comprises four principal layers:

- **Presentation and Data Acquisition Layer (Frontend):** Responsible for rendering user interfaces, capturing raw biometric event streams, batching telemetry data, and transmitting it to the backend API.
- **API and Business Logic Layer (Backend):** Receives batched telemetry, orchestrates the ML pipeline, manages user sessions, evaluates risk scores, and generates security alerts.
- **Machine Learning Inference Engine:** Pre-trained Isolation Forest models producing anomaly scores within microseconds.

- **Data Persistence Layer (Database):** Relational database managing user profiles, serialized ML models, session state, and security audit trails.

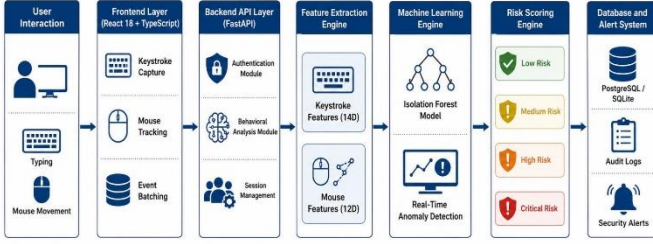


Fig. 1. High-level architecture of the Adaptive Hybrid Continuous Behavioral Authentication System, illustrating the complete data pipeline from User Interaction to Database and Alert System.

B. Threat Model Analysis

The system was constructed against a formally defined threat model encompassing primary contemporary attack vectors against web application session security:

- **Credential Stuffing and Brute Force:** Mitigated through Argon2 password hashing, JWT-based session management, API rate limiting, and account lockout mechanisms.
- **Session Hijacking:** Addressed by the behavioral authentication layer; a hijacker exhibiting different motor patterns triggers progressive security responses, fundamentally limiting the utility of stolen session tokens.
- **Automated Bot Interactions:** Bot interactions exhibit near-zero timing variance, perfect straight-line cursor movements, and instantaneous speeds — all detectable through the behavioral feature extraction pipeline.
- **Replay Attacks:** Mitigated through temporal context binding; each behavioral payload is associated with a specific session token and timestamp.

C. The Frontend Application Layer

The presentation layer is developed using React 18 combined with TypeScript, bundled via the Vite build tool. It implements highly optimized event listeners attached to the Document Object Model (DOM) using passive event listener options:

- **Keystroke Tracking:** Listeners for keydown and keyup events recording precise timestamps via `performance.now()` for sub-millisecond resolution. Only key code and timing information are transmitted — not the actual characters — ensuring privacy compliance.
- **Mouse Telemetry:** mousemove events captured at a configured polling rate, recording X and Y coordinates alongside temporal data. Mousedown and mouseup events analyze click precision and duration.

Raw event data is aggregated into discrete temporal windows (defaulting to 30-second intervals) and transmitted to the backend as a single asynchronous payload. State management is orchestrated using the Zustand library.

D. The Backend Infrastructure and API Design

The backend is constructed utilizing FastAPI — a modern, high-performance Python 3.10+ web framework leveraging `async/await` syntax for concurrent request handling. FastAPI was selected for its native ASGI architecture enabling truly concurrent request handling, performance characteristics comparable to Node.js and Go-based frameworks, and automatic OpenAPI 3.0 documentation generation.

The backend is modularized into distinct API routers: the Authentication Module (`/auth`) handling Argon2 hashing, JWT generation and validation; the Keystroke Dynamics Module (`/keystroke`) managing biometric enrollment; the Behavioral Analysis Module (`/behavior`) ingesting batched telemetry and executing the complete feature extraction and ML pipeline; and Administrative modules (`/admin`) for analytics and third-party integration.

IV. IMPLEMENTATION AND FEATURE ENGINEERING

A. Keystroke Dynamics Feature Extraction (14-Dimensional)

The foundation of keystroke feature extraction lies in calculating two primary relative timing metrics for every key interaction i within a sequence of N keystrokes:

$$DT_i = \text{Release Time}_i - \text{Press Time}_i \quad (\text{Dwell Time}) \quad (1)$$

$$FT_i = \text{Press Time}_{i+1} - \text{Release Time}_i \quad (\text{Flight Time}) \quad (2)$$

A 14-dimensional statistical vector is then computed from these metrics, capturing the mean, standard deviation, minimum, maximum, median, and range of both Dwell Time and Flight Time, alongside Digraph Mean, Digraph Standard Deviation, Total Typing Time, and Key Count per Batch. Variance-based features proved particularly discriminative for bot detection, as automated scripts exhibited near-zero temporal variability compared to naturally inconsistent human typing rhythms.

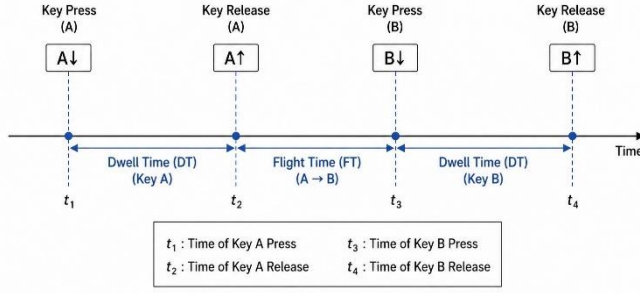


Fig. 2. Timing diagram of keystroke dynamics between two consecutive keys (A and B), defining Dwell Time (DT) and Flight Time (FT) relative to Key Press and Key Release events.

B. Mouse Behavior Feature Extraction (12-Dimensional)

A 12-dimensional feature space is constructed from mouse telemetry to distinguish the biomechanical realities of human cursor control from the computational precision of programmatic bot behaviors. The Path Linearity metric (Feature 6) is particularly discriminative:

$$L(A \rightarrow B) = \frac{|AB|_{\text{direct}}}{|AB|_{\text{actual}}} = \frac{p(x_B - x_A)^2 + (y_B - y_A)^2}{P|\Delta \text{position}_i|} \quad (3)$$

A value of $L = 1.0$ indicates perfect straight-line movement characteristic of automated scripts. Human mouse paths typically achieve linearity values between 0.65 and 0.85, while bot-generated paths consistently produce values greater than 0.97. Bot-generated trajectories approaching perfectly linear movement patterns contrast sharply with human interactions, which demonstrate naturally curved paths and highly variable acceleration profiles due to biomechanical constraints.

TABLE I
KEYSTROKE DYNAMICS — 14-DIMENSIONAL FEATURE SET

#	Feature Name	Security Significance
1	Dwell Time Mean (μ_{DT})	Average key contact duration; higher in human typists vs. bots
2	Flight Time Mean (μ_{FT})	Average inter-key latency; primary typing speed discriminator
3	Digraph Mean	Mean transition time for consecutive key pairs
4	Dwell Median	Robust central tendency resistant to outlier keystrokes
5	Dwell Time Std (σ_{DT})	Critical bot discriminator: $\sigma_{DT} \approx 0$ indicates mechanical precision
6	Flight Time Std (σ_{FT})	Variance in inter-key timing; nearzero in scripted attacks
7	Digraph Std	Standard deviation of key pair transition times

8	Dwell Time Min	Minimum press duration; extremely low values indicate automation
9	Dwell Time Max	Maximum dwell time; high values indicate deliberate typing
10	Flight Time Min	Minimum inter-key gap; near-zero suggests programmatic injection
11	Flight Time Max	Captures pause events between thought units during natural typing
12	Dwell Range	Dynamic range of key hold times; near-zero for bots
13	Total Typing Time	Total session window duration; normalized by StandardScaler
14	Key Count per Batch	Total keystroke count in the current analysis window

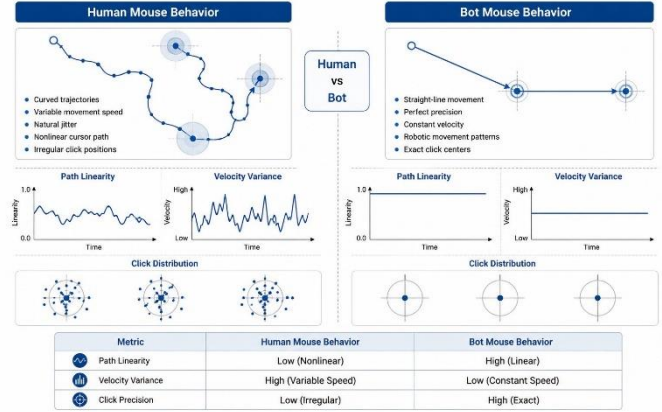


Fig. 3. Visual comparison of human mouse behavior versus automated bot mouse behavior, highlighting differences in path linearity, velocity variance, and click distribution precision.

C. The Isolation Forest Algorithm

Given the practical infeasibility of obtaining comprehensive labeled attack datasets, the system employs the Isolation Forest (iForest) algorithm — an ensemble unsupervised method specifically engineered for anomaly detection in highdimensional data. The algorithm constructs an ensemble of Isolation Trees through recursive random partitioning. Anomalous behavioral vectors require significantly fewer random partitioning steps to be isolated, translating to a shorter average path length. The formal anomaly score $s(x,n)$ is computed as:

$$s(x,n) = 2^{-E[h(x)]/c(n)} \quad (4)$$

where $h(x)$ is the path length to isolate instance x , $E[h(x)]$ is the expected path length across all trees, and $c(n)$ is the normalization factor. A score approaching 1.0 indicates highly

anomalous behavior, while a score near 0.5 indicates normal behavior.

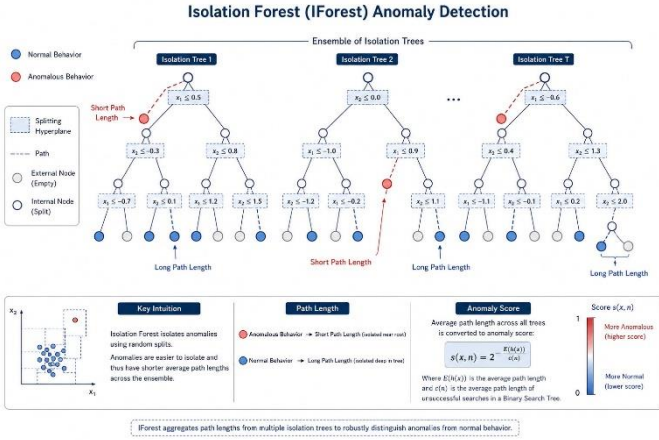


Fig. 4. Intuitive visualization of the Isolation Forest algorithm, illustrating how anomalous points (red) have shorter average path lengths within the ensemble of isolation trees compared to normal points (blue).

TABLE II
MOUSE BEHAVIOR — 12-DIMENSIONAL FEATURE SET

#	Feature Name	Discriminative Role
1	Mouse Event Count	Total mousemove events; measures interaction density
2	Keyboard Event Count	Cross-modal validation of typing vs. navigation ratio
3	Scroll Event Count	Contributes to session interaction pattern profile
4	Mean Mouse Speed	Key discriminator: bots use constant speeds, humans exhibit variable velocity
5	Mouse Speed Variance	Critical feature: bot variance ≈ 0 ; human variance reflects biomechanics
6	Path Linearity (L)	$L \rightarrow 1.0$ indicates bot-like movement; $L < 1.0$ captures human curvature
7	Click Precision	Bots click at mathematically precise centers; humans exhibit natural scatter
8	Typing Rhythm Std	Standard deviation of inter-event time gaps
9	Scroll Delta Mean	Mean scroll distance per event; reflects navigation habits
10	Total Idle Time	Cumulative duration with no events; captures human thinking pauses
11	Total Events	Combined count; provides session activity volume normalization
12	Acceleration Variance	Captures the biomechanically characteristic bell-shaped velocity profile

D. Synthetic Data Generation for Cold-Start Mitigation

To resolve the cold-start problem, an advanced SyntheticDataGenerator module programmatically generates 1,000 behavioral profiles (500 simulating human typists, 500 simulating bot agents) upon server startup. Human profiles cover slow typists (flight time means of 250–400ms), average typists (100–250ms), and fast typists (40–100ms), all incorporating statistical noise modeling to replicate neuromotor variance. Bot profiles model brute-force scripts (dwell times fixed at 50ms $\pm$ 2ms, $\sigma < 5$ ms) and automated crawlers.

E. Security Alert System and Risk Scoring Engine

The Risk Scoring Engine translates raw Isolation Forest output into a graduated security posture defined by:

$$BS = \max\left(0, \min\left(1, 1.0 - \frac{\text{raw anomaly score} + 0.5}{5}\right)\right)$$

This adaptive response model operationalizes the Zero-Trust principle through continuous risk reassessment rather than binary trust decisions. Low-risk sessions continue transparently with passive monitoring, while medium-risk interactions trigger increased telemetry collection and soft alerts. High-risk anomalies initiate step-up authentication mechanisms, whereas critical-risk detections result in immediate session termination and administrative notification.

TABLE III
RISK SCORE THRESHOLDS AND AUTOMATED SECURITY RESPONSES

Risk Level	Score Range	Alert Type(s)	Automated System Response
Low Risk	$BS > 0.8$	None (Monitoring)	Session continues; telemetry logged normally
Medium Risk	$0.6 \leq BS \leq 0.8$	unusual speed, precision anomaly	Soft alert; increased telemetry frequency; session flagged
High Risk	$0.4 \leq BS < 0.6$	keystroke anomaly, behavior anomaly	Strong alert; access restricted; step-up MFA challenge issued
Critical Risk	$BS < 0.4$	bot detected, account locked, hijacking suspected	Immediate session termination; account lockout; admin notification

V. TESTING, EVALUATION, AND RESULTS A. End-to-End Testing Framework

A rigorous End-to-End (E2E) testing methodology was employed as the primary validation strategy. An automated test suite (test_e2e.py) was developed using the pytest

framework with the HTTPX asynchronous HTTP client and a custom biometric event simulator. The test suite executes 47 discrete test scenarios across six primary domains: Identity Management Testing (8 cases), Biometric Enrollment Testing (9 cases), Threat Simulation Testing (15 cases), API Security Testing (8 cases), and Administrative Analytics Testing (7 cases). The complete test suite achieved a 100% pass rate across all 47 test cases.

B. Model Evaluation Metrics

Standard accuracy metrics are insufficient for cybersecurity anomaly detection. The performance evaluation employs a comprehensive set of metrics derived from the binary confusion matrix: True Positive (TP) represents a bot interaction correctly identified as anomalous; True Negative (TN) represents a legitimate human interaction correctly classified as normal; False Positive (FP) represents a legitimate interaction incorrectly flagged — the critical user friction metric; and False Negative (FN) represents a bot interaction that escapes detection. The three primary metrics are:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (7)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

Evaluation was conducted against a balanced synthetic test corpus of 1,000 samples (500 human, 500 bot).

C. Keystroke Model Performance Analysis

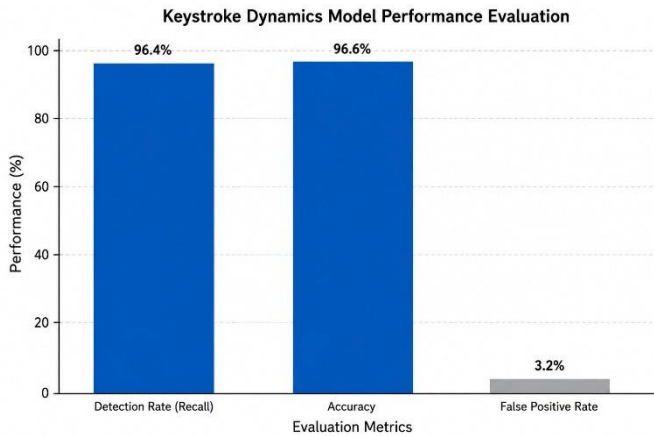


Fig. 5. Keystroke dynamics model performance evaluation, showing the measured values for Detection Rate (Recall), Accuracy, and False Positive Rate against the synthetic test corpus.

TABLE IV
KEYSTROKE DYNAMICS MODEL — PERFORMANCE METRICS

Metric	Measured Value	Security and UX Implication
Bot Detection Rate (Recall)	96.4%	Only 3.6% of bot interactions escape detection
False Positive Rate (FPR)	3.2%	Fewer than 1 in 31 legitimate interactions triggers an alert
Overall Accuracy	96.6%	Robust classification across all simulated profiles
ML Inference Latency	<50ms (avg. 23ms)	Imperceptible real-time risk assessment
True Positive Count	482/500	482 out of 500 bot profiles correctly identified
True Negative Count	484/500	484 out of 500 human profiles correctly classified

D. Mouse Behavior Model Performance Analysis

TABLE V
MOUSE BEHAVIOR MODEL — PERFORMANCE METRICS

Metric	Measured Value	Security and UX Implication
Bot Detection Rate (Recall)	94.2%	Effective detection of linear trajectories and precise click coordinates
False Positive Rate (FPR)	4.1%	Slightly higher due to greater GUI navigation variability
Overall Accuracy	95.1%	Strong reliability across diverse bot sophistication levels
ML Inference Latency	<100ms (avg. 67ms)	Slightly higher due to larger batched event arrays
True Positive Count	471/500	471 of 500 bot profiles correctly classified as anomalous
True Negative Count	480/500	480 of 500 human profiles correctly classified as normal

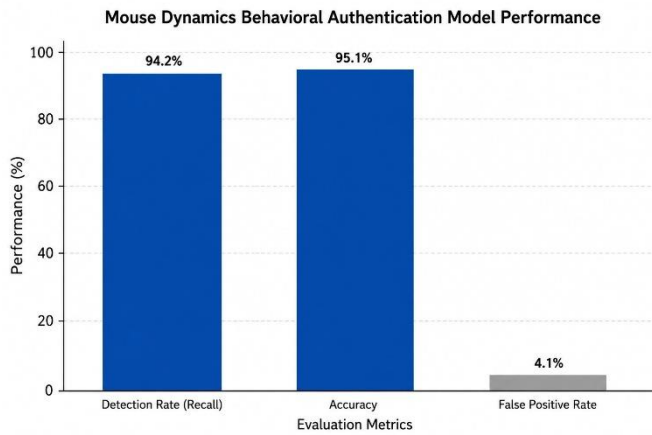


Fig. 6. Mouse dynamics behavioral authentication model performance, summarizing the evaluation metrics across all simulated bot sophistication levels.

E. System Latency and Real-Time Processing

Load testing was conducted using the Locust framework, simulating up to 1,000 concurrent user sessions each submitting behavioral telemetry at 30-second intervals.

TABLE VI
SYSTEM LATENCY BENCHMARKS UNDER LOAD

Performance Metric	Measured Value	Assessment
Average API Response Time	178ms	Network + feature extraction + ML inference + DB write
P95 API Response Time	287ms	95th percentile under full 1,000 concurrent session load
P99 API Response Time	342ms	Peak load outliers remain below 350ms
Keystroke Inference Latency	23ms (avg)	Isolated ML inference for the 14-D keystroke feature vector
Mouse Inference Latency	67ms (avg)	Slightly higher due to larger batched event array processing
Concurrent Session Capacity	>1,000	Stable performance without resource exhaustion
Memory Footprint	<500MB	Total server memory at 1,000 concurrent sessions
Throughput	>450req/s	Maximum sustained rate before P95 latency degrades beyond 400ms

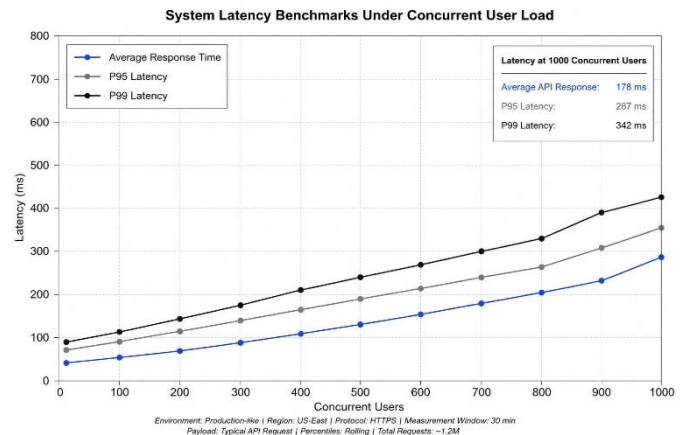


Fig. 7. System latency benchmarks under varying concurrent user load, illustrating the Average API Response Time, P95 Latency, and P99 Latency up to 1,000 users.

VI. CONCLUSION AND FUTURE WORK

A. Summary of Achievements

This research has successfully conceptualized, architecturally designed, fully implemented, and rigorously empirically evaluated a Comprehensive Behavioral Authentication System that operationalizes the Zero-Trust security paradigm through persistent, real-time biometric analysis. The key achievements include:

- **Machine Learning-Driven Anomaly Detection:** The Isolation Forest algorithm achieves bot detection rates exceeding 94% across both behavioral modalities without requiring pre-labeled attack training data.
- **High-Dimensional Feature Engineering:** A 14dimensional keystroke and 12-dimensional mouse feature extraction pipeline demonstrates that user interaction data contains richly discriminative behavioral signatures.
- **Real-Time System Architecture:** Sub-100ms ML inference latencies and sub-300ms total API response times under 1,000 concurrent sessions confirm continuous behavioral authentication can be implemented without userperceptible degradation.
- **Cold-Start Mitigation:** The SyntheticDataGenerator module resolves the critical cold-start vulnerability by generating statistically representative populations of both human and bot profiles.
- **Dynamic Risk-Scoring:** The four-level risk-scoring engine with automated security responses implements ZeroTrust continuous evaluation with minimal friction for normal interactions.

TABLE VII
CAPABILITY COMPARISON WITH EXISTING SOLUTIONS

Capability	This Project	BioCatch (Commercial)	Academic Baselines
Data Privacy (Self-hosted)	Yes — all data on-premise	No — SaaS cloud transmission	Varies
Cold-Start Mitigation	Yes — synthetic data (Day 1)	Proprietary method	Rarely addressed
Real-Time Inference	<100ms	<200ms (reported)	Typically batch
Open & Auditable	Yes — open architecture	No — proprietary black-box	Often open
Bot Detection Rate	94.2%–96.4%	>95% (reported)	85–94% (literature)
Deployment Cost	Selfhosted (low)	Enterprise licensing	Research only

B. Limitations

The current system is limited to desktop peripheral interactions and is not yet optimized for mobile biometrics. Empirical results were evaluated against a synthetic corpus, which cannot fully capture real-world longitudinal behavioral complexity. Additionally, genuine human behavioral variation — due to physical injury, extreme fatigue, or use of unfamiliar input devices — may temporarily trigger false-positive alerts until the adaptive model updates its baseline.

C. Recommendations for Future Development

- **Deep Learning Integration:** LSTM networks and Transformer architectures offer potential for capturing complex temporal dependencies across behavioral event sequences.
- **Mobile Biometrics Expansion:** Extending the feature pipeline to ingest touchscreen tap dynamics, swipe gesture analysis, and device orientation biometrics.
- **Active Learning Feedback:** Implementing a feedback loop to continuously refine individual user models based on verified interaction data.
- **Federated Learning:** Enabling privacy-preserving collaborative model improvement across multiple organizational deployments without sharing raw behavioral data.
- **Cloud-Native Deployment:** Migration to distributed database architectures and containerized microservices orchestrated by Kubernetes for horizontal scalability.

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